

Power Supply Systems

Part II

Phase 3: Kelvin's Law and Economic Size of Conductor

Aim: To understand Kelvin's law from absolute zero and become ready for semester proof and numerical questions.

Style: Simple English, SN ma'am style, clean graph, no overlapping labels, formula after physical meaning.

Target: Statement, proof, graph, limitations, numerical method, economic cross-section and economic current density.

Subject: Power Supply Systems

Section: Part II

Phase: 3 – Kelvin's Law

Level: Absolute zero to semester-question-ready

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1 Phase 3 Roadmap

Phase 3 Goal

In Phase 2, we learnt about conductors and comparison of supply systems. Now we ask one very important economic question:

What should be the most economical size of conductor?

Kelvin's law gives the answer.

Conductor size → Capital cost → Energy loss cost → Total annual cost → Minimum cost → Kelvin's law

Memory Hook

Small conductor is cheap but causes high I^2R loss. Large conductor has low loss but high initial cost. Kelvin's law finds the best middle point.

2 Why Economic Conductor Size is Needed

A transmission or distribution line needs conductor material. If we choose a very small conductor:

- initial cost is low,
- resistance is high,
- I^2R loss is high,
- voltage drop is high,
- cost of wasted energy becomes high.

If we choose a very large conductor:

- resistance is low,
- I^2R loss is low,
- but initial cost is very high.

So, there must be one conductor size where the total annual cost is minimum.

Definition

The **economic size of conductor** is that conductor size for which the total annual cost of the line becomes minimum.

3 Two Opposite Costs in a Line

Kelvin's law is based on two opposite costs.

3.1 Annual Capital Charge

This is the annual cost due to money spent on conductor and line construction. It includes mainly:

- interest on capital cost,

- depreciation,
- maintenance-related annual charges.

If conductor area increases, conductor cost increases.

Area increases $\Rightarrow$ Capital charge increases

3.2 Annual Cost of Energy Loss

Energy is lost due to conductor resistance.

$$P_{\text{loss}} = I^2 R$$

But,

$$R = \frac{\rho l}{a}$$

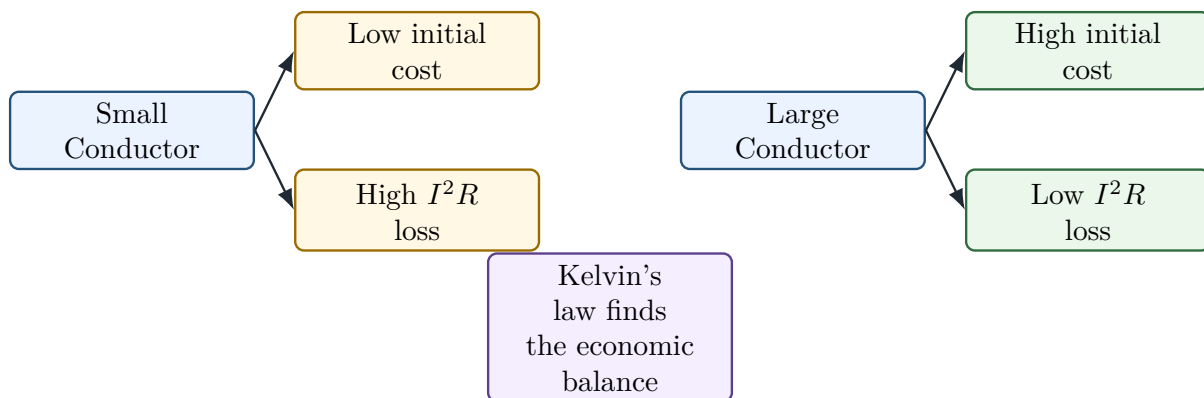
So,

$$R \propto \frac{1}{a}$$

Therefore,

Area increases $\Rightarrow$ Resistance decreases $\Rightarrow$ Loss cost decreases

3.3 Cost Balance Diagram



4 Statement of Kelvin's Law

Definition

Kelvin's Law: The most economical cross-section of a conductor is obtained when the variable part of the annual capital charge is equal to the annual cost of energy wasted in the conductor.

In simple mathematical form:

Variable annual capital charge = Annual cost of energy loss

Memory Hook

Kelvin's law means:

$$\text{Extra money spent on conductor} = \text{Money saved by reducing loss}$$

5 Derivation of Kelvin's Law

5.1 Assumptions and Symbols

Let,

a = area of cross-section of conductor

P_1 = constant part of annual charge

P_2a = variable annual charge proportional to conductor area

Therefore,

$$\text{Annual capital charge} = P_1 + P_2a$$

Now resistance of conductor is:

$$R = \frac{\rho l}{a}$$

Energy loss is due to resistance. So annual cost of energy wasted is inversely proportional to conductor area.

$$\text{Annual cost of energy wasted} = \frac{P_3}{a}$$

where P_3 is a constant.

5.2 Total Annual Cost

Total annual cost C is:

C = annual capital charge + annual cost of energy loss

$$C = P_1 + P_2a + \frac{P_3}{a}$$

For minimum cost:

$$\frac{dC}{da} = 0$$

$$\frac{d}{da} \left(P_1 + P_2a + \frac{P_3}{a} \right) = 0$$

$$0 + P_2 - \frac{P_3}{a^2} = 0$$

$$P_2 = \frac{P_3}{a^2}$$

Multiplying both sides by a ,

$$P_2 a = \frac{P_3}{a}$$

But,

$$P_2 a = \text{variable annual capital charge}$$

and,

$$\frac{P_3}{a} = \text{annual cost of energy wasted}$$

Therefore,

$$\text{Variable annual capital charge} = \text{Annual cost of energy wasted}$$

Hence Kelvin's law is proved.

Semester Answer Style

For a 4–6 marks proof:

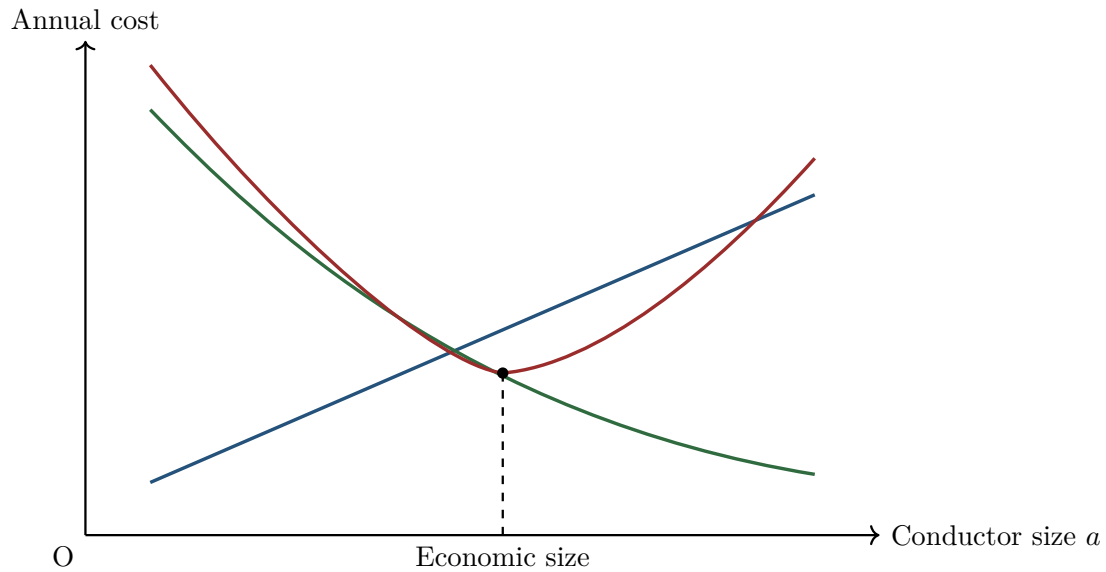
1. Write economic size definition.
2. Write Kelvin's law statement.
3. Write $C = P_1 + P_2 a + \frac{P_3}{a}$.
4. Differentiate with respect to a .
5. Put $\frac{dC}{da} = 0$.
6. Get $P_2 a = \frac{P_3}{a}$.
7. Explain the final equation in words.

6 Graphical Representation of Kelvin's Law

The graph contains three curves:

1. Capital charge curve: increases with conductor size.
2. Energy-loss cost curve: decreases with conductor size.
3. Total annual cost curve: has a minimum point.

6.1 Clean Graph



6.2 Legend for the Graph

Curve	Meaning
— Capital charge curve	Increases with conductor size because more conductor material is used.
— Energy-loss cost curve	Decreases with conductor size because resistance decreases.
— Total annual cost curve	First decreases, then increases. Its lowest point gives economic conductor size.

Semester Answer Style

In exam, do not put too many words inside the graph. Draw the graph cleanly and write the explanation below it.

7 Limitations of Kelvin's Law

Kelvin's law is very useful for understanding economy, but it cannot be applied exactly in practical cases.

1. It is difficult to estimate the actual load curve for future years.
2. The annual cost of energy loss cannot be predicted exactly.
3. The assumption that capital cost varies directly with conductor area is not fully correct.
4. It does not consider mechanical strength of the conductor.
5. It does not consider safe current density.
6. It does not consider voltage drop limitation.
7. It does not consider corona loss.
8. Standard conductor sizes are available in market, not every calculated size.
9. Interest and depreciation rates may change.

10. Future load growth is not accurately known.

Semester Answer Style

For a 3 marks question, write any five limitations. For a 5 marks question, write seven to eight limitations with one-line explanation.

8 Economic Current Density

Definition

Economic current density is the current carried per unit economic cross-sectional area of the conductor.

If I is the current and a is the economic area,

$$\delta = \frac{I}{a}$$

where,

δ = economic current density

If I is in ampere and a is in cm^2 , then:

$$\delta = \text{A}/\text{cm}^2$$

Semester Answer Style

If the question asks for economic current density, first find the economic area using Kelvin's law. Then use:

$$\delta = \frac{I}{a}$$

9 General Numerical Solving Method

For Kelvin's law numerical, use this fixed method.

1. Write given data.
2. Calculate line current.
3. Write resistance in terms of area:

$$R = \frac{rl}{a}$$

4. Calculate copper loss:

$$P_{\text{loss}} = nI^2R$$

5. Calculate annual energy loss in kWh.
6. Convert annual energy loss into rupees.
7. Calculate variable annual capital charge.
8. Apply Kelvin's law:

$$\text{Variable annual capital charge} = \text{Annual cost of energy loss}$$

9. Solve for a .
10. If asked, find:

$$\delta = \frac{I}{a}$$

Memory Hook

Numerical flow:

$$I \rightarrow R \rightarrow I^2 R \rightarrow \text{kWh} \rightarrow \text{Rs.} \rightarrow a$$

10 Useful Formula Derivation for Constant Current

Let,

n = number of conductors carrying current

I = line current

r = resistance of one conductor for given length and unit area

a = area of conductor

Then,

$$R = \frac{r}{a}$$

Copper loss is:

$$P_{\text{loss}} = nI^2 R$$

$$P_{\text{loss}} = nI^2 \frac{r}{a}$$

If line operates for H hours per year, annual energy loss is:

$$E_{\text{loss}} = \frac{nI^2 r H}{1000a}$$

If cost of energy is C_e Rs/kWh, then annual cost of energy loss is:

$$C_{\text{loss}} = C_e \frac{nI^2 r H}{1000a}$$

Let variable annual capital charge be:

$$C_{\text{capital}} = Ka$$

By Kelvin's law:

$$Ka = C_e \frac{nI^2 r H}{1000a}$$

$$a^2 = \frac{C_e n I^2 r H}{1000 K}$$

$$a = \sqrt{\frac{C_e n I^2 r H}{1000 K}}$$

Common Mistake

This formula is only a shortcut. In semester answers, the stepwise method is safer and more accepted.

11 Solved Example 1: Basic Cable Type Numerical

11.1 Problem

A two-conductor cable carries a constant current of 200 A. The cost of cable is Rs. $(50a + 25)$ per metre where a is the area in cm^2 . Interest and depreciation is 10%. The cable works for 6000 hours per year. Cost of energy is Rs. 0.05 per kWh. Resistance of one conductor per metre is:

$$R = \frac{1.85 \times 10^{-4}}{a} \Omega$$

Find the most economical cross-section.

11.2 Solution

Given,

$$I = 200 \text{ A}$$

For two conductors, total resistance per metre is:

$$R_{\text{total}} = 2 \times \frac{1.85 \times 10^{-4}}{a}$$

$$R_{\text{total}} = \frac{3.70 \times 10^{-4}}{a} \Omega$$

Power loss per metre:

$$P_{\text{loss}} = I^2 R$$

$$P_{\text{loss}} = 200^2 \times \frac{3.70 \times 10^{-4}}{a}$$

$$P_{\text{loss}} = \frac{14.8}{a} \text{ W}$$

Annual energy loss:

$$E_{\text{loss}} = \frac{14.8}{a} \times \frac{6000}{1000}$$

$$E_{\text{loss}} = \frac{88.8}{a} \text{ kWh}$$

Annual cost of energy loss:

$$C_{\text{loss}} = 0.05 \times \frac{88.8}{a}$$

$$C_{\text{loss}} = \frac{4.44}{a} \text{ Rs}$$

Variable part of capital cost:

$$50a \text{ Rs}$$

Annual capital charge:

$$C_{\text{capital}} = 10\% \text{ of } 50a$$

$$C_{\text{capital}} = 5a$$

By Kelvin's law:

$$5a = \frac{4.44}{a}$$

$$a^2 = \frac{4.44}{5}$$

$$a^2 = 0.888$$

$$a = 0.942 \text{ cm}^2$$

12 Solved Example 2: Three-Phase Line with Load Cycle

12.1 Problem

A three-phase, 33 kV overhead line supplies:

2500 kW for 8 h/day

1500 kW for 10 h/day

1000 kW for 6 h/day

Power factor is unity. The line is used for 365 days. Line cost is Rs. $(16250 + 5000a)$ per km, where a is in cm^2 . Length of line is 8 km. Resistance of 1 km conductor of area 1 cm^2 is 0.176Ω . Energy cost is Rs. 0.05 per kWh. Interest and depreciation is 8%. Find economic cross-section.

12.2 Step 1: Calculate Currents

For three-phase system:

$$P = \sqrt{3}VI \cos \phi$$

Since power factor is unity,

$$I = \frac{P}{\sqrt{3}V}$$

For 2500 kW:

$$I_1 = \frac{2500 \times 10^3}{\sqrt{3} \times 33 \times 10^3}$$

$$I_1 = 43.74 \text{ A}$$

For 1500 kW:

$$I_2 = \frac{1500 \times 10^3}{\sqrt{3} \times 33 \times 10^3}$$

$$I_2 = 26.24 \text{ A}$$

For 1000 kW:

$$I_3 = \frac{1000 \times 10^3}{\sqrt{3} \times 33 \times 10^3}$$

$$I_3 = 17.50 \text{ A}$$

12.3 Step 2: Resistance of One Conductor

For 8 km length:

$$R = \frac{0.176 \times 8}{a}$$

$$R = \frac{1.408}{a} \Omega$$

12.4 Step 3: Annual Energy Loss

For three-phase line:

$$P_{\text{loss}} = 3I^2R$$

Annual energy loss:

$$E = \frac{3R}{1000} \times 365 \left[I_1^2 \times 8 + I_2^2 \times 10 + I_3^2 \times 6 \right]$$

Substituting values:

$$E = \frac{37045.66}{a} \text{ kWh}$$

Annual cost of energy loss:

$$C_{\text{loss}} = 0.05 \times \frac{37045.66}{a}$$

$$C_{\text{loss}} = \frac{1852.28}{a} \text{ Rs}$$

12.5 Step 4: Annual Capital Charge

Variable cost per km:

$$5000a$$

For 8 km:

$$5000a \times 8 = 40000a$$

Annual interest and depreciation:

$$C_{\text{capital}} = 8\% \text{ of } 40000a$$

$$C_{\text{capital}} = 3200a$$

12.6 Step 5: Apply Kelvin's Law

$$3200a = \frac{1852.28}{a}$$

$$a^2 = \frac{1852.28}{3200}$$

$$a^2 = 0.5788$$

$$\boxed{a = 0.761 \text{ cm}^2}$$

13 Solved Example 3: Economic Current Density

13.1 Problem

A three-phase 11 kV cable supplies 1500 kW at 0.8 power factor lagging. It is used for 300 days in a year at 8 hours per day. Resistance of 1 km length of each conductor of 1 cm² area is 0.173Ω. Cost of energy is Rs. 0.02 per kWh. Interest and depreciation is 12%. Capital cost per km is:

$$\text{Rs. } (8000 + 20000a)$$

Find most economical current density.

13.2 Step 1: Current

$$P = \sqrt{3}VI \cos \phi$$

$$I = \frac{1500 \times 10^3}{\sqrt{3} \times 11 \times 10^3 \times 0.8}$$

$$I = 98.41 \text{ A}$$

13.3 Step 2: Operating Hours

$$H = 300 \times 8$$

$$H = 2400 \text{ h}$$

13.4 Step 3: Resistance

For 1 km:

$$R = \frac{0.173}{a} \Omega$$

13.5 Step 4: Annual Energy Loss

$$E = \frac{3I^2RH}{1000}$$

$$E = \frac{3(98.41)^2(0.173)(2400)}{1000a}$$

$$E = \frac{12063.53}{a} \text{ kWh}$$

Annual cost of loss:

$$C_{\text{loss}} = 0.02 \times \frac{12063.53}{a}$$

$$C_{\text{loss}} = \frac{241.27}{a} \text{ Rs}$$

13.6 Step 5: Annual Capital Charge

Variable capital cost:

$$20000a$$

Annual charge:

$$C_{\text{capital}} = 12\% \text{ of } 20000a$$

$$C_{\text{capital}} = 2400a$$

13.7 Step 6: Apply Kelvin's Law

$$2400a = \frac{241.27}{a}$$

$$a^2 = 0.10053$$

$$a = 0.317 \text{ cm}^2$$

13.8 Step 7: Economic Current Density

$$\delta = \frac{I}{a}$$

$$\delta = \frac{98.41}{0.317}$$

$$\delta = 310.4 \text{ A/cm}^2$$

Common Mistake

In economic current density problems, do not stop after finding a . The final answer must be $\delta = I/a$.

14 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
Economics of feeder design	Economic conductor size	State and prove Kelvin's law.
Conductor size	Balance between cost and loss	Draw cost curve and explain economical size.
Energy loss	I^2R loss and cost of wasted energy	Find economic cross-section from load cycle.
Current density	Current per unit area	Find most economical current density.
Limitations	Practical difficulties	Mention limitations of Kelvin's law.

15 Most Predictable Semester Questions from Phase 3

No.	Question	Marks
1	State and prove Kelvin's law for economic size of conductor.	4–6
2	Draw and explain the graphical representation of Kelvin's law.	4
3	Mention the limitations of Kelvin's law.	3–5
4	A conductor cost is given as $(A + Ba)$. Find economic cross-section.	7–8

5	Apply Kelvin's law to a three-phase overhead line with daily load cycle.	8
6	Determine most economical current density in a three-phase cable.	7
7	Explain why too small and too large conductor sizes are uneconomical.	3–4
8	Derive the condition: Variable annual capital charge = Annual cost of energy loss	4

16 Exam Answer Templates

16.1 Template 1: State and Prove Kelvin's Law

Semester Answer Style

Answer sequence:

1. Define economic conductor size.
2. State Kelvin's law.
3. Write annual capital charge $P_1 + P_2a$.
4. Write annual loss cost $\frac{P_3}{a}$.
5. Write total annual cost:

$$C = P_1 + P_2a + \frac{P_3}{a}$$

6. Differentiate and put $\frac{dC}{da} = 0$.

7. Get:

$$P_2a = \frac{P_3}{a}$$

8. Write final meaning in words.

16.2 Template 2: Kelvin's Law Numerical

Semester Answer Style

Answer sequence:

1. Given data.
2. Current calculation.
3. Resistance in terms of a .
4. I^2R loss.
5. Annual energy loss.
6. Annual cost of energy loss.
7. Variable annual capital charge.
8. Apply Kelvin's law.
9. Find a .
10. Find current density if asked.

16.3 Template 3: Limitations

Semester Answer Style

Answer sequence:

1. Start: Kelvin's law is theoretical.
2. Write load curve difficulty.
3. Write capital charge assumption issue.
4. Write mechanical strength and safe current density.
5. Write corona and voltage drop.
6. Write standard conductor sizes and future load growth.

17 Phase 3 Final Memory Sheet

One-Page Memory Sheet

1. Economic conductor size gives minimum total annual cost.
2. Small conductor: low capital cost but high I^2R loss.
3. Large conductor: low I^2R loss but high capital cost.
4. Kelvin's law balances these two costs.
5. Statement:

$$\text{Variable annual capital charge} = \text{Annual cost of energy loss}$$

6. Total annual cost:

$$C = P_1 + P_2a + \frac{P_3}{a}$$

7. Minimum condition:

$$\frac{dC}{da} = 0$$

8. Final result:

$$P_2a = \frac{P_3}{a}$$

9. Economic current density:

$$\delta = \frac{I}{a}$$

10. Numerical method:

$$I \rightarrow R \rightarrow I^2R \rightarrow \text{kWh} \rightarrow \text{Rs.} \rightarrow a$$

18 Phase 3 Self-Test

Answer these without seeing notes:

1. What is economic size of conductor?
2. State Kelvin's law.
3. Why does capital charge increase with conductor size?
4. Why does energy-loss cost decrease with conductor size?
5. Write total annual cost equation.
6. Derive Kelvin's law from $C = P_1 + P_2a + \frac{P_3}{a}$.
7. Draw the Kelvin's law cost curve.
8. Mention five limitations of Kelvin's law.
9. What is economic current density?
10. What is the stepwise method for a Kelvin's law numerical?

Common Mistake

Do not move to distributor numericals until you can prove Kelvin's law without seeing notes. This topic is a repeated semester weapon.

End of Phase 3

Next Phase:

Phase 4 – Distributor Fed at One End

Current-loading diagram, voltage-drop diagram, concentrated loading, uniformly distributed loading and voltage at load points.